THE UNIVERSITY OF MELBOURNE

ACCT90002 Financial Statement Analysis

Week 4: Accounting analysis (2)

Accounting analysis – [Weeks 3-4]

- It aims to evaluate the degree to which a firm's accounting captures the underlying business economics (week3)
- It enables analysts to assess the degree of distortion in a firm's accounting numbers (week3 & 4)
- It enables the analyst to recast accounting numbers to create unbiased accounting numbers (week4)
- Sound accounting analysis improves the reliability of conclusions from financial analysis

Lecture outline

1. Income statement analysis

2 . Balance sheet analysis

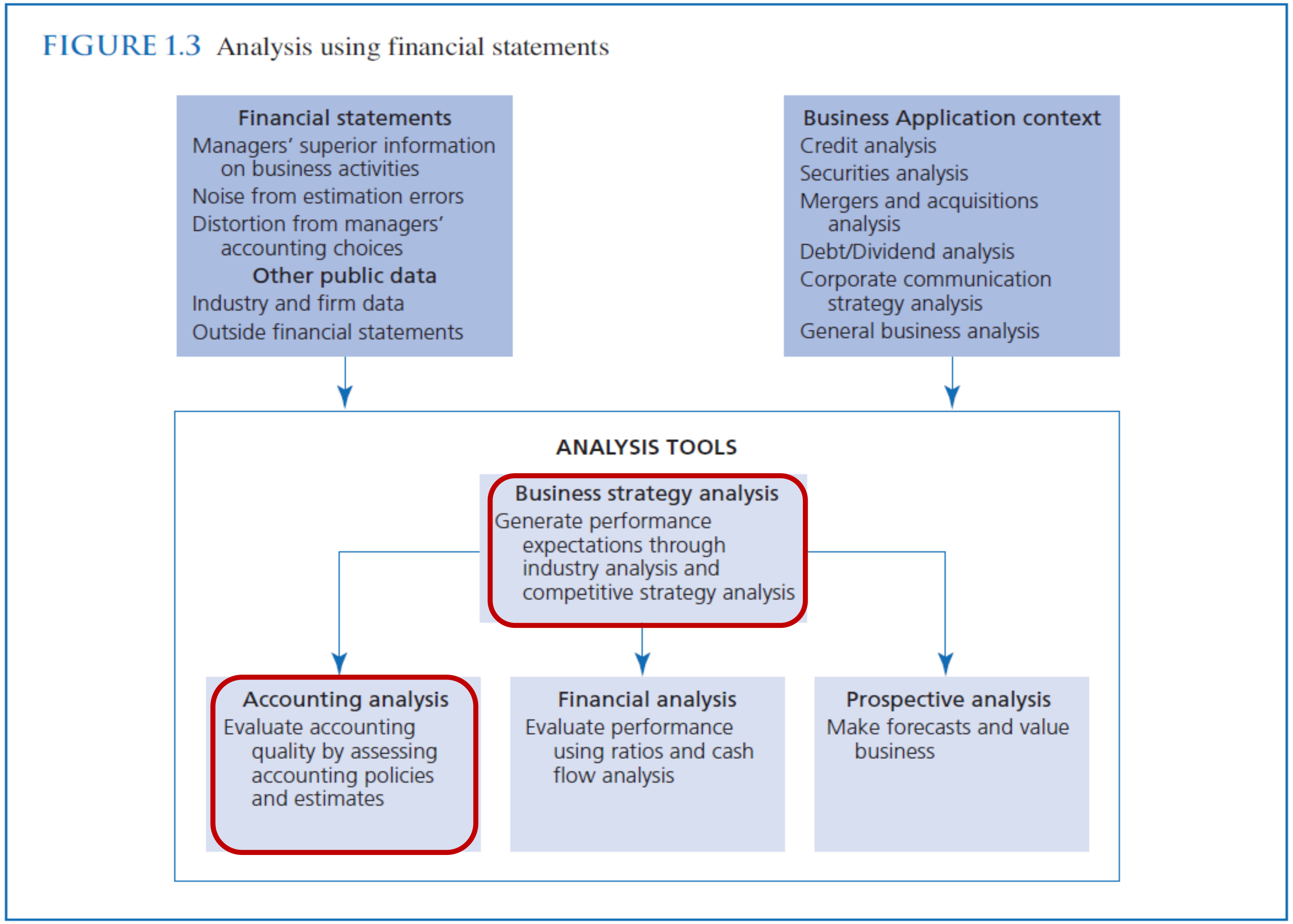
   1) Asset distortions

   2) Liability distortions

   3) Equity distortions

3. Recasting accounting numbers

Textbook chapters 3-4



Adjustments and statement analysis

- There are times when an analyst should perform adjustments to financial statements to:
  - Reflect financial statements more consistently with economic reality
  - Reflect financial statements more consistently with the past
  - Reflect financial statements more consistently with other companies
- Doing so ensures an “apples v apples” comparison, and prevents misinterpretation

Income Statement Analysis

Income statement analysis

The income statement provides information about the **profitability** of a firm for a **period of time**:

Revenues/Sales	X
Cost of sales	..
Gross profit	..
Selling, general and administrative expenses	..
Depreciation/amortization	..
Operating profit (EBIT)	..
Interest income/expense	..
Income before taxes	..
Income taxes	..
Net income	Y

Income statement analysis

–**Revenues/sales**: value inflows from selling activities

–**Gross profit**: value creation (net of direct outflows) from selling activity

–**Operating profit**: value creation from running the operations (includes value changes from other operating factors that are not direct selling activities

–**Profit before tax**: Overall value creation from running the company (does not include range of valuation adjustments to assets/liabilities) before corporate tax

–**Net profit after tax**: Overall value creation from running the company (does not include range of valuation adjustments to assets/liabilities) after corporate tax

–**EPS**: Overall value creation from running the company (does not include range of valuation adjustments to assets/liabilities) after corporate tax, on a per share basis

Income statement analysis

- Distortions primarily arise due to the following:
  - Early or delayed recognition of revenue

    Consider associated receivables (asset, early) or unearned revenue (liability, late)
  - Early or delayed recognition of expenses

    Consider associated provision (liability, early) or the absence of recording.
  - Changing presentation (not technically a distortion)

Revenue recognition

- **Revenue** is one of the most significant line items in a firm's financial statements

- Managers typically have the best information on the uncertainties associated with revenue recognition

- But: managers may also have incentives to **accelerate** the recognition of revenues to manage earnings upwards

  → **overstated** receivables and earnings

Balance Sheet Analysis

Assets = Liabilities + Equity

Assets

Assets:

Resources that a firm **owns or controls** as a result of past business transactions, and which are expected to produce **future economic benefits** that can be **measured** with **a reasonable** degree of certainty.

Distortions arise because of ambiguity about:

1. Who **owns/controls** the economic resources?

   Joint ventures etc. SPEs

2. Are the economic resources likely to provide **future economic benefits** that can be **measured** with **reasonable certainty**?

3. Fair value relative to book value

4. Recognition: intangible

Issues with control

- Why is control an issue:
  - Affects the way a subsidiary is accounted for:

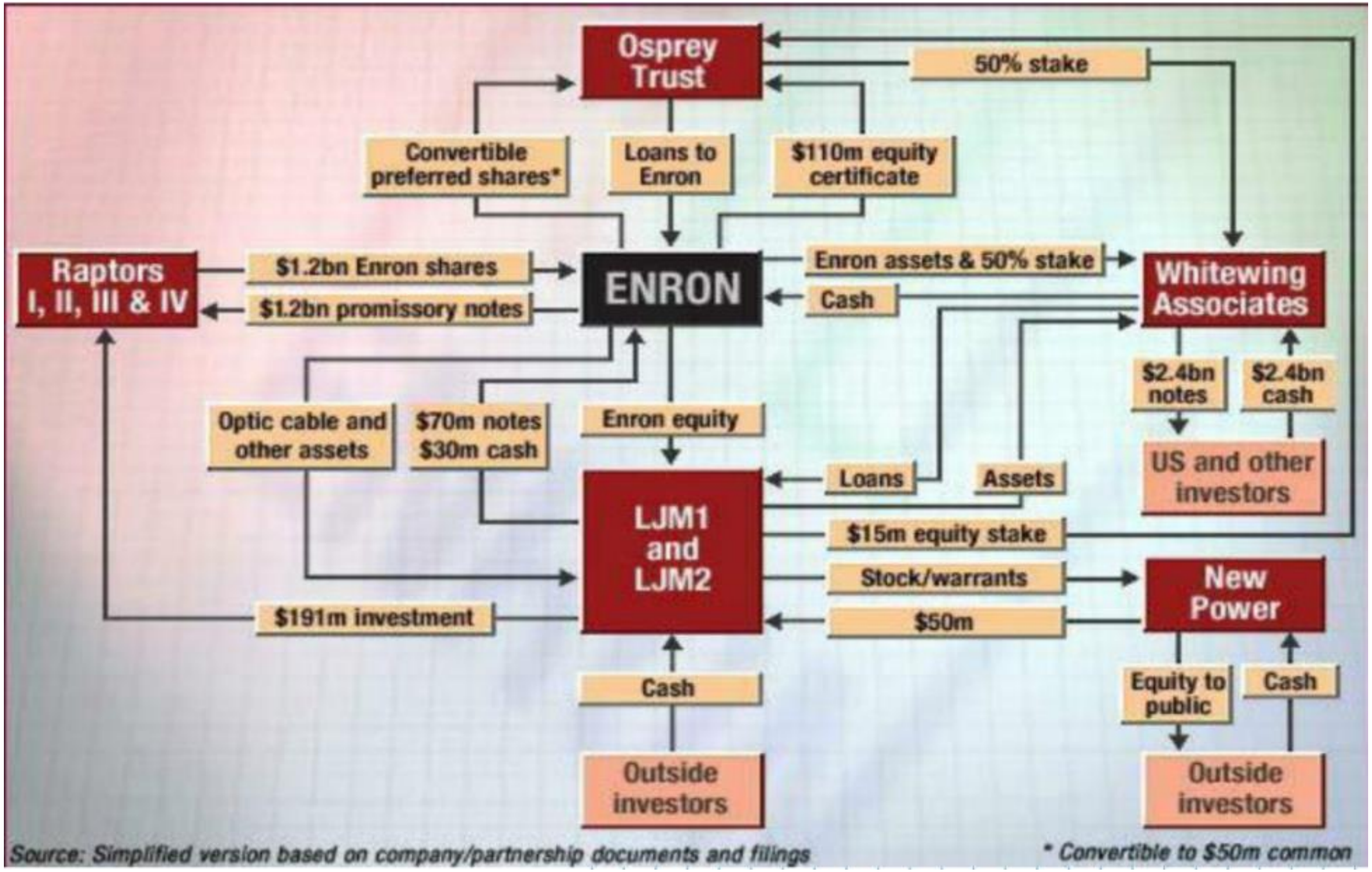
    (a) investment;

    (b) equity-accounting; or

    (c) full consolidation

- Potentially lead to vastly different outcomes in financial analysis

Enron and its SPEs



Source: Liu, Q., & Vasarhelyi, M. A. (2014). Big questions in AIS research: Measurement, information processing, data analysis, and reporting. Journal of information systems, 28(1), 1-17.

Issues with valuation

- Why is valuation an issue:
  - ➢ Affects the value at which an asset is recorded. These values have further consequences.
  - – Example accounting using cost-method: Based on price paid
  - – Example accounting using fair value method: Based on "market" price
- Other issues related to value:
  - – Depreciation and amortisation
  - – Impairment

Issues with fair value

Issues with fair value:

- How to determine fair value?
- How to deal with inefficient/illiquid markets?

Take, for example, the disclosures required in financial statements about assets:

- Quoted (easy to value for accounting purposes)
- Unquoted, observable inputs (easier, but not easy to value for accounting purposes)
- Unobservable inputs (harder to value for accounting purposes)

Assets valuation

	Fair value measurements							
	Quoted price in active markets (Level 1)		Using observable inputs (Level 2)		Using unobservable inputs (Level 3)		Total	
	2024 $m	2023 $m	2024 $m	2023 $m	2024 $m	2023 $m	2024 $m	2023 $m
Assets								
Cash and cash equivalents (measured at fair value)	-	-	37,255	27,566	-	-	37,255	27,566
Trading assets[1]	31,507	26,388	14,233	10,614	15	2	45,755	37,004
Derivative financial instruments[1]	131	935	54,214	59,448	25	23	54,370	60,406
Investment securities[1]	111,060	71,356	21,055	16,924	1,343	1,397	133,458	89,677
Net loans and advances	-	-	24,429	21,159	357	729	24,786	21,888
Total	142,698	98,679	151,186	135,711	1,740	2,151	295,624	236,541
Liabilities								
Deposits and other borrowings (designated at fair value)	-	-	43,001	33,889	-	-	43,001	33,889
Derivative financial instruments[1]	393	218	54,846	57,241	15	23	55,254	57,482
Payables and other liabilities	5,804	4,841	219	426	-	-	6,023	5,267
Debt issuances (designated at fair value)	-	-	1,816	1,336	-	-	1,816	1,336
Total	6,197	5,059	99,882	92,892	15	23	106,094	97,974

Issues with impairment and allowance

- **Issues with impairment and allowance**
  - How to determine whether impairment has occurred?
  - How to value asset (especially given often non-tradeable)?
  - How to determine allowance sufficient?
- Signs of impairment for **non-current assets**:
  - declining non-current asset turnover and lapsing of key assumptions;
  - write-downs by other firms in same industry;
  - overpayment or poor integration of key acquisitions
- Allowances and write-downs of **current assets** (Accounts receivables and loan receivables; inventory)
  - ➢ **Warning signs:** growing days receivables; business downturns for major clients; growing loan defaults; growing days' inventory, inventory write-downs by competitors

Intangible assets

- Some firms' most important assets are **excluded** from the balance sheet:

  1. R&D investments
  2. Software development outlays
  3. Brands and membership bases created through advertising and promotions. **(internally generated goodwill)**

- Accounting rules prescribe the expensing of intangible outlays because of the **uncertainty** about their future economic value.

Consider 2 companies, both investing in their productive technology: (1) capital equipment; (2) R&D. Assume they both have same monetisation opportunities, and same chance of success. Same accounting?

Liabilities

Liabilities:

Economic obligations arising from benefits received in the past, and for which the amount and timing is known with reasonable certainty.

Distortions arise because of **ambiguity** about:

1. Has an obligation really been incurred?
2. Can the obligation be ***measured*** reliably?

Examples:

– Unearned revenues

– Provisions and contingent liabilities

– Post-employment benefit obligations

Liability distortions

- **Understatements** of liabilities occur when:

1. Firm has **key commitments** that are difficult to value and are therefore not considered liabilities for financial reporting purposes.
2. Managers have incentives to overstate the soundness of the firm’s financial position or boost reported earnings.

→ Understated **leverage**

Dealing with distortions

- Some distortions very difficult (or impossible) for an analyst to fully "correct"
- General approaches to distortions:
  - Identify but do not address
  - Identify and remove distorting item completely (i.e. earnings "before distorting item")
  - Identify and **recast**
- Recasting financial statements is a technical accounting task

Recasting Financial Statements

Why make accounting adjustments?

Accounting standards

- Financial statements may have been prepared for tax purposes
- Accounting rules may not capture all the subtleties arising from transactions and events
- Accounting measurement used to record assets may not reflect the economics of the firm

Accounting choices within GAAP

- If the target firm employs different accounting methods from those of the comparison firms, the analyst must decide whether to make accounting adjustments

Management bias

- Management has considerable input into, and discretion over, the financial reporting process, including selecting accounting polices and making accounting estimates
- Management may distort financial reports to increase earnings to meet analysts' forecast

Recasting – General Approach

- Recasting involves ”correcting” or introducing accounting items to bring the financial statements into line with:
  - A standard formulation that can be compared between companies or across time
  - The underlying economic reality

- Basic approach:
  - Recasting follows the same rules as regular double entry accounting
  - Adjustments made to correct assets and liabilities
  - Adjustments made to earnings in the current period affect earnings items in current period
  - Adjustments made to earnings in previous periods affect retained earnings
  - Each period, adjust all periods required until distortion no longer exists/relevant
  - Recasted assets will have to be appropriately impaired/amortised
  - Use present values where time duration is appropriate (expected value of liability/asset)

Recasting Financial Statements

Firms can vary in the terminology and formats used to report financial results.

Differences in financial statement format, classification and terminology make it difficult to compare performance across firms and for the same firm over time.

Balance sheets, income statements and statements of cash flows may be recast into a common format.

Design a standardised template for the balance sheet, income statement and cash flow statement that can be used for any firm.

Adjustment for management bias

Some distortions that have been identified in the literature are:

- Provisions
- Asset impairment
- Timing of revenue recognition
- Expense capitalisation

Adjustment for management bias

Provisions

- The most common earnings-management approach (133 out of 515 cases) is to recognise too much or too little in **provisions**.
- Involve **judgment** and **estimation**.
- Managers have the best information regarding the probability and estimated amount of the obligation. But they may also have the incentives and discretion to distort provisions information.

> - Provisions refer to liabilities or potential losses that a company anticipates but for which the exact timing or amount is uncertain. Provisions are made to account for future expenses or losses that are likely to occur, but their exact timing or amount cannot be determined with certainty at the present time

Adjustment for management bias

Provisions

- In 2017, NPL was sued in a $100 product liability case
- NPL chose not to recognise
- But then NPL paid $50 in 2018 and $50 in 2019 to settle the case
- Assume an analyst successfully predict these two actual payments
- Please make the accounting adjustments accordingly (assume the tax rate is 30%):
- Discount rate is 5% for any cash flows.

What is the effect on the financial statements?

– Liabilities understated in 2017 (by two payments), and 2018 (by one payment). Equity the opposite.

– Profits overstated in 2017 and 2018 by litigation settlement (after tax)

Recasting Provisions

Addressing the issue in 2017, with two payments ahead…

– Is it in the future? If so, present values…

– Does it have a tax consequence? If so, then address tax also

Expected value of liability
= PV of cash flows out
= 50/(1.05^2) + 50/1.05
= 45.4 + 47.6
= 93

Expected value of tax effect
= 93 x 30%
= 27.9 ~ 28

Recasting Provisions

Addressing the issue in 2018, with one payment ahead…

Expected value of liability
= PV of cash flows out
= 50/1.05
= 47.6 ~ 48

Expected value of tax effect
= 47.6 x 30%
= 14.3 ~ 14

For the economic reality:

1. Net liability in 2017 = +65
2. Net Liability in 2018 = +34 (Prov 48 – DTL 14)
3. Difference = adjustment to profit= +31
4. New profit should result in $81

Recording Steps:

1. Record 2017 entry
2. Reverse 2017 entry now in 2018
3. Record new 2018 entry (including tax effect)

Adjustment for management bias

Provisions

- In 2017, NPL was sued in a $100 product liability case
- NPL chose not to recognise
- But then NPL paid $50 in 2018 and $50 in 2019 to settle the case
- Assume an analyst successfully predict these two actual payments
- Please make the accounting adjustments accordingly (assume the tax rate is 30%):

FIGURE 4.7 Tax adjustment for NPL and its impact on two ratios

2017	Assets	=	Liabilities		+	Equity	
			Provisions	Other		Profit	Other
Reported	1 100			500		100	500
1 Provision			93			−93	
2 Deferred tax				−28		28	
Adjusted	1 100		93	472		35	500

Ratios	Reported	Adjusted
Return on assets	9.1%	3.2%
Debt to assets	45.5%	51.4%

PV in 2017 of the $50 paid in 2018 (5% discount rate): $48

PV in 2017 of the $50 paid in 2019 (5% discount rate): $45

Total: $93

FIGURE 4.8 The 2018 adjustments for NPL

2018	Assets	=	Liabilities		+	Equity	
			Provisions	Other		Profit	Other
Reported	1 050			500		50	500
3 Opening			93	−28			−65
4 Reverse			−93	28		65	
5 Provision			48			−48	
6 Deferred tax				−14		14	
Adjusted	1 050		48	486		81	435

Ratios	Reported	Adjusted
Return on assets	4.8%	7.7%
Debt to assets	47.6%	50.9%

- Opening backlog adjustment to account for prior year adjustments.
  - Apply prior year adjustments to RE
  - Reverse them to Profit
- Apply new adjustments:
  - PV in 2018 of the $50 paid in 2019 (5% discount rate): **$48**

Adjustment for management bias

Asset impairment

- The second most common earnings management approach relates to recognising too much or too little **~~asset impairment~~** of property plant and equipment, investments and intangibles.
- Involve **judgment** and **estimation**, because the second-hand market for non-current assets is typically incomplete and illiquid.
- Managers have the best information regarding the fair values for such assets. But they may also have the incentives and discretion to distort asset impairment information.

> - An asset is impaired if its fair value falls below its carrying amount.
> - IAS 36 requires a firm to recognise impairment as it occurs.

Adjustment for management bias

Asset impairment

In 2016, Morgan's Orchard Limited acquired Kidd's Gala Limited, resulting in a goodwill of $200.

In 2019, Morgan's Orchard publicly admitted that it had paid too much for Kidd's Gala and announced that a goodwill impairment will be recorded.

Assume an analyst successfully predict this goodwill impairment in 2017, and want to record a goodwill write-down of $100.

Please make the accounting adjustments accordingly:

What is the effect on the financial statements?

– Assets overstated; equity overstated

– Profits overstated in 2017

Adjustment for management bias

Asset impairment

Addressing the issue in **2017**…

FIGURE 4.10 Asset impairment adjustment for Morgan's Orchard in 2017

2017	Assets		=	Liabilities	+	Equity	
	Goodwill	Other				Profit	Other
Reported	200	900		500		100	500
1. Impairment	−100			1		−100	
Adjusted	100	900		500		0	500

reduce the asset (goodwill) and reduce profit (goodwill impairment expense)

Ratios	Reported	Adjusted
Return on assets	9.1%	0.0%
Debt to assets	45.5%	50.0%

2

Addressing the issue in **2018**…

- No need for further transactions as no change in reported accounts and no change in adjusted

FIGURE 4.11 Adjustment for Morgan's Orchard in 2018

2018	Assets		=	Liabilities	+	Equity	
	Goodwill	Other				Profit	Other
Reported	200	900		500		100	500
2. Opening	−100			1			−100
Adjusted	100	900		500		100	400

- Opening backlog adjustment: reduce opening goodwill and reduce RE

Ratios	Reported	Adjusted
Return on assets	9.1%	10.1%
Debt to assets	45.5%	50.0%

2

Adjustment for management bias

Asset impairment

Addressing the issue in **2019**…

• Need to make additional entry to reverse the affect in current period reported earnings as recorded already in 2017 (now retained earnings)

FIGURE 4.12 Adjustment for Morgan's Orchard in 2019

2019	Assets		=	Liabilities	+	Equity	
	Goodwill	Other				Profit	Other
Reported	50	900		500		−50	500
3…Opening	−100						−100
4…Reversal	100					100	
Adjusted	50	900		500		50	400

1

Opening backlog adjustment reduce opening goodwill and RE

2

reverse them to Profit

Ratios	Reported	Adjusted
Return on assets	−5.3%	5.3%
Debt to assets	52.6%	52.6%

3

Adjustment for management bias

Timing of revenue recognition

- One of the most significant line items in a firm's financial statements
- Involve **judgment** and **estimation**, because …….
- Managers have the best information, but they may also have the incentives to accelerate (or delay) the recognition of revenues to manage earnings upwards (or downwards for a cookie jar strategy)

Adjustment for management bias

Timing of revenue recognition

In 2018 Impatient Imports Limited (IIL) wrongly recognized sales when goods left the depot (it was not until the client received the goods, up to a week later, that ownership passed to them).

In 2019, the new auditors uncovered the fault and IIL reported that it will restate its financial statements.

Assume an analyst successfully detected this issue in 2018, and decided to make the following accounting adjustments:

- reduce sales revenue by $220.
- an inventory reversal of $10 related to these sales
- a deferred tax effect of $69. (tax rate is 30%)

What is the effect on financial statements?

– Revenues / profits overstated – and accordingly, accounts receivable overstated
– Cost of goods sold overstated – and accordingly, Inventory understated
– Tax consequences as the transactions impacts taxable profits

Adjustment for management bias

Timing of revenue recognition

In **2019**, when transaction is actually recorded, need to offset and then restate

FIGURE 4.13 Adjustment for IIL in 2018

2018	Assets	=	Liabilities	+	Equity		
					Revenue	Expenses	Other
Reported	1 217		569		1 430	−1 282	500
1 Receivables/revenue	−220				−220		
2 Inventories/cost of sales	10					10	
3 Deferred tax			−69			69	
Adjusted	1 007		500		1 210	−1 203	500

Ratios	Reported	Adjusted
Return on sales	10.3%	0.6%
Return on assets	12.2%	0.7%
Debt to assets	46.8%	49.7%

FIGURE 4.14 Adjustment for IIL, 2019

2019	Assets	=	Liabilities	+	Equity		
					Revenue	Expenses	Other
Reported	1 167		500		1 480	−1 313	500
Adjustments							
Opening	−210		−69				−141
Reversal	210		69		210	−69	
Adjusted	1 167		500		1 690	−1 382	359

Opening backlog adjustment

Reversal adjustment

Ratios	Reported	Adjusted
Return on sales	11.3%	18.1%
Return on assets	14.3%	26.4%
Debt to assets	42.8%	42.8%

Adjustment for management bias

Expense capitalisation

- Some firms' most important assets are excluded from the balance sheet:
  1. R&D investments
  2. Software development outlays
  3. Brands and membership bases created through advertising and promotions.
- Accounting rules prescribe the expensing of intangible outlays because of the uncertainty about their future economic value. But this leads to:

  1. Inflated/deflated return on assets and return on equity (possibly) difficult for analyst when forecasting

  2. More difficult for the analyst to assess whether business model works: mismatch between revenues and expenses.

Adjustment for management bias

Expense capitalisation

In 2017, Capital City Properties Limited (CCPL) capitalised $50 of renovation expenses on a newly acquired property and depreciated over five years. Other firms in the industry expense all renovation expenses.

An analyst noted this difference and decided to unwind this renovation expenses capitalization.

Please make the accounting adjustments accordingly:

What are the effects?

– Assets overstated

– Expenses understated

– Need to account for tax

Adjustment for management bias

Expense capitalisation

Reversing the effect in **2017**

– Reverse the capitalisation of $40
– Account for 30% deferred taxation to address the tax affect

FIGURE 4.15 Adjustment for CCPL in 2017

2017	Asset		=	Liabilities	+	Equity	
	Capitalised	Other				Profit	Other
Reported	40	1 100		512		128	500
1 Capitalisation	−40					−40	
2 Deferred taxation				−12		12	
Adjusted	0	1 100		500		100	500

Ratios	Reported	Adjusted
Return on assets	11.2%	9.1%
Debt to assets	44.9%	45.5%

Reversing the effect in **2018**

– Reverse the capitalisation of $40 and tax in prior period
– Back out the depreciation/ amortization that is being charged by the company in reported earnings ($10)

FIGURE 4.16 Adjustment for CCPL in 2018

2018	Asset		=	Liabilities	+	Equity	
	Capitalised	Other				Profit	Other
Reported	30	1 072		509		93	500
Opening	−40			−12			−28
Depreciation	10					10	
Deferred taxation				3		−3	
Adjusted	0	1 072		500		100	472

Opening backlog adjustment
Depreciation adjustment
Defer tax adjustment

Ratios	Reported	Adjusted
Return on assets	8.4%	9.3%
Debt to assets	46.2%	46.6%

Key Takeaways

- Income statements and Balance Sheets analysis: finding distortions

- Recasting financial statements is an important step to facilitate comparability among financial statements analysed.

- Analysts should focus on evaluating and adjusting accounting measures that describe the firm’s key strategic value drivers.

- It is important to keep in mind that many accounting adjustments will be estimates.

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See you at the seminar